

- Document Control
- System Name: Application
- Version:
- Document Owner:
- Status:
- Last Updated:
- Reviewers:

- Executive Summary
- The Application will process customer and order source CSV data by loading it into Delta tables, performing data transformations, maintaining historical customer and order records, and generating summary and aggregate spend tables leveraging Delta Live Tables.

- Business Objectives
- Enable automated ingestion and cleansing of customer and order data.
- Maintain accurate historical records according to Slowly Changing Dimension (SCD) Type 2 methodology.
- Provide insights into customer aggregate spend through regular summary tables.

- In-Scope
- Loading customer and order data from CSV sources to Delta tables.
- Data schema definition and enforcement.
- Calculating "TotalAmount" for orders.
- Data cleaning (null/duplicate removal).
- Creating and updating the order summary SCD Type 2 table.
- Aggregating and reporting customer spend.
- Implementation with Delta Live Tables.

- Out-of-Scope
- Data sources outside customer and order CSV files.
- Real-time ingestion from external systems not specified.
- Reporting and dashboard creation beyond aggregate tables.

- Stakeholders
- Data Engineering Team
- Business Analysts
- Operations Team

- Business Requirements

BR-APP-001 — Ingest Customer Data

- Description: Read customer CSV data from defined volume location and load into Delta table.
- Business Value: Standardizes customer data storage for analytics.
- Priority: High

BR-APP-002 — Ingest Order Data

- Description: Read order CSV data from defined volume location and load into Delta table.
- Business Value: Ensures order transactions are available for downstream processing.
- Priority: High

BR-APP-003 — Define Schemas

- Description: Enforce customer (CustId, Name, EmailId, Region) and order (OrderId, ItemName, PricePerUnit, Qty, Date, CustId) table schemas.
- Business Value: Maintains data integrity and consistent structure.
- Priority: High

BR-APP-004 — Calculate TotalAmount

- Description: Add "TotalAmount" column to order table as product of PricePerUnit and Qty.
- Business Value: Enables spend analytics and reporting.
- Priority: Medium

BR-APP-005 — Cleanse Source Data

- Description: Remove null and duplicate records from both customer and order tables.
- Business Value: Ensures dataset quality and reliability.
- Priority: High

BR-APP-006 — Create OrderSummary Table

- Description: Create "ordersummary" Delta table if not exists in specified catalog and schema.
- Business Value: Centralizes enriched order-customer joined data for analytics.
- Priority: High

BR-APP-007 — Implement SCD Type 2

- Description: Maintain and update historical customer and order information (active/inactive, StartDate, EndDate).
- Business Value: Supports auditing and longitudinal customer reporting.
- Priority: High

BR-APP-008 — Create CustomerAggregateSpend Table

- Description: Create "customeraggregatespend" table with Name, TotalAmount, and Date fields.
- Business Value: Measures and reports on total customer spend per period.
- Priority: Medium

BR-APP-009 — Aggregate Spend Calculations

- Description: Aggregate TotalAmount from ordersummary by grouping on Name and Date, loading results into customeraggregatespend.
- Business Value: Supports business intelligence with up-to-date spend metrics.
- Priority: Medium

BR-APP-010 — Use Delta Live Tables

- Description: Implement all steps via Delta Live Tables framework.
- Business Value: Enables automated pipeline management and reliability.
- Priority: High
- Assumptions & Constraints
 - Source CSV files are available and accessible at defined volume paths.
 - Customer and order schemas are consistent and stable.
 - Delta Lake infrastructure and permissions are in place.
- Success Metrics
 - 100% customer and order records processed without critical errors.
 - "ordersummary" and "customeraggregatespend" tables complete and updated per data load.
 - SCD history correctness verified in test loads.
- Risks & Mitigations
 - Source data quality issues → Mitigation: Implement strict cleansing logic.
 - Schema changes in source files → Mitigation: Monitor for schema drift, enforce schema validation.
 - Pipeline execution failures → Mitigation: Use Delta Live Tables for resiliency.
- Approval
- Business Owner:
- IT/Data Owner:
- Date:
- Signatures: